An organization may have different types of data, such as business-related data (invoices, orders, customer information, etc.) and operational information (web server and database server logs, application logs, application metrics, etc.). Bundling all sorts of data into a single cluster is not unusual, but it may not be a best practice. It may be a better strategy to create multiple clusters for varied data shapes, with customized configurations for each cluster. For example, a business-critical data cluster may be running on an on-premises cluster with greater memory and disk space options configured, while an application-monitoring data cluster has a slightly different setup.

Additional node on a personal machine

If you are running a node on your personal laptop or PC, you can instantiate an additional node from the same installation folder. Rerun the shell script by passing two additional parameters: the `path.data` and `path.logs` options, pointing to the respective directories, as shown here:

```
$>cd <INSTALL_DIR>/bin
$>./elasticsearch -Epath.data=../data2 -Epath.logs=../log2
```

This command spins up an additional node with the given data and log folders. If you now issue the `GET _cat/nodes` command, a second node should appear in the list.

Each node needs to work with specific responsibilities at times. Some may need to worry about data-related activities such as indexing and caching, while others may be required to coordinate a client's requests and responses. Some tasks may also include node-to-node communications and cluster-level management. To enable such responsibilities, Elasticsearch has a set of roles for nodes to undertake a specific set of responsibilities when assigned to that role. We discuss node roles in the next section.

NODE ROLES

Every node plays multiple roles, from being a coordinator to managing data to becoming a master. The developers of Elasticsearch keep fiddling with nodes, so you may see new roles appear now and then. Table 3.2 lists the node roles that we can categorize.

Table 3.2 Node roles and responsibilities

Role	Description
Master node	Its primary responsibility is cluster management.
Data node	Responsible for document persistence and retrieval.
Ingest node	Responsible for the transformation of data via pipeline ingestion before indexing.
Machine learning node	Handles machine learning jobs and requests.
Transform node	Handles transformation requests.
Coordination node	This is the default role. It takes care of incoming client requests.